

4(a)	any three from: 1 donor not needed ; 2 immediate effect ; 3 <i>idea that</i> easy to administer treatment ; 4 no immune response / no rejection ; 5 less invasive ; 6 AVP ; e.g. lower risk to health / cheaper / quicker	3
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Question	Answer	Marks
4(b)	<i>any four from:</i> 1 given a drug to increase number of stem cells (in bone marrow) ; 2 <i>ref. to</i> virus / vector, containing, normal / healthy, allele ; 3 remove, bone marrow / stem cells ; 4 mix stem cells with, viral / vector (to allow transfer of normal allele) ; 5 radiotherapy / drug, to make space in bone marrow / to kill stem cells (in bone marrow) ; 6 (transduced stem) cells, infused / injected, into blood ; 7 (lymphocytes) produce functioning ADA ; 8 AVP ; e.g. (gamma) retrovirus / adeno-associated virus e.g. tissue from bone marrow, purified / sorted, to obtain stem cells e.g. cells are grown in culture to check the ADA gene is active	4

Question	Answer	Marks
4(c)	any three from: 1 expensive ; 2 cure / long term treatment / no longer chronically ill / better quality of life ; 3 no need for regular, injections / treatments or only a single treatment ; 4 cultural / religious, objections ; 5 no donor needed ; 6 <i>ref.</i> more money available to health system in the long term ; 7 may cause cancer ; 8 stressful ;	3